

SOFTWARE DESIGN DOCUMENT

MARO-v2

Multi-Agent Radiology Orchestrator

Sovereign Medical Imaging Intelligence System (SMIIS)

Version 2.0 — February 2026
AUTHORITATIVE SOURCE OF TRUTH

Target Audience:

Machine Learning Engineers • Data Scientists • Clinicians • DevOps Engineers

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I. Clinical Mandate

1.1 Problem Statement: The Fragmented Patient Journey

The current state of medical imaging interpretation is characterized by systemic fragmentation that endangers patient outcomes and burdens clinical workflows. When a patient presents with symptoms requiring imaging analysis, they encounter a cascade of disconnected systems: imaging archives (PACS), electronic health records (EHR), laboratory information systems (LIS), and specialist consultations. Each system operates in isolation, forcing clinicians to mentally synthesize information across interfaces while maintaining temporal context.

1.1.1 The Fragmentation Topology

Spatial Fragmentation	CT/MRI slices viewed in isolation; 3D relationships lost	RoMedFormer 3D-RoPE embedding
Temporal Fragmentation	Prior studies not automatically correlated; growth patterns missed	Temporal GraphRAG with Neo4j
Semantic Fragmentation	Findings described without grounding to pixels; audit impossible	Chain-of-Box spatial anchoring
Workflow Fragmentation	AI outputs as "black boxes"; no correction mechanism	LangGraph Time-Travel checkpointing

1.2 The "Glance Effect": Primary Adversary

The Glance Effect describes the phenomenon where Vision-Language Models (VLMs) arrive at statistically correct diagnoses through superficial pattern matching rather than anatomically-grounded reasoning. This occurs because Supervised Fine-Tuning (SFT) optimizes for output mimicry, not reasoning fidelity.

1.2.1 Glance Effect Manifestation

Definition: A model exhibits the Glance Effect when it produces a correct diagnostic label while attending to incorrect anatomical regions, non-diagnostic features, or dataset artifacts.

Example: A model trained on chest X-rays correctly predicts "Pneumonia" but attention mapping reveals it focused on the cardiac silhouette border (a common imaging artifact) rather than the consolidation in the lung parenchyma. The prediction is "right for the wrong reason."

1.2.2 Clinical Consequence Matrix

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Spatial Misattribution	Labels "RLL Nodule" while looking at liver	Missed malignancy	IoU Spatial Reward (R_spatial)
Temporal Blindness	Calls "New Finding" that existed 2 years prior	Unnecessary biopsy	GraphRAG longitudinal query
Artifact Confounding	Misreads metal artifact as pathology	False positive cascade	RoMedFormer artifact inpainting
Confidence Miscalibration	High confidence on ambiguous findings	Delayed specialist referral	GRPO advantage normalization

1.3 Design Mandate

MARO-v2 is designed to transform the diagnostic pipeline from a "Black Box" to a "Glass Box" system. Every diagnostic conclusion must be:

- Spatially Grounded: Linked to explicit voxel coordinates via Chain-of-Box
- Temporally Contextualized: Compared against the patient's longitudinal record
- Reasoning-Transparent: Derived through auditable <think> traces
- Human-Correctable: Modifiable via Time-Travel state rollback
- Sovereignly Deployed: Executed entirely on-premise with no cloud dependencies

II. Modular Agent Mesh

2.1 Architectural Philosophy: Division of Cognitive Labor

Medical image interpretation is not a singular cognitive task. It comprises multiple sequential operations that vary in computational complexity, latency tolerance, and error criticality. MARO-v2 distributes these operations across specialized agents, each optimized for a specific cognitive function.

2.1.1 The Three-Layer Architecture

INGESTION	Data normalization, knowledge graph population	Batch (seconds)	Rule-based + Embedding	High (correctable)
TRIAGE	Key slice selection, relevance scoring	Real-time (ms)	Lightweight ViT (~12M params)	Medium (recoverable)
ANALYSIS	Deep reasoning, diagnosis generation, validation	Interactive (seconds)	Large VLM (70B+ params)	Zero (critical path)

2.2 Layer 1: Ingestion Layer

2.2.1 Agent 1: Medical Librarian (Document & Metadata Ingestion)

Purpose: Transform raw, disorganized medical data into a structured, queryable knowledge representation.

Input Sources: DICOM files, PDF radiology reports, clinical notes, lab results

Output Artifacts: Knowledge Graph nodes, vector embeddings, temporal relationships

Data Schema — Ingestion Event:

event_id	UUID	Unique identifier for ingestion event	System-generated
timestamp	ISO8601	Time of ingestion	System clock
source_type	Enum	DICOM PDF HL7_FHIR FREE_TEXT	File extension / header
study_instance_uid	String	DICOM Study UID for cross-reference	DICOM header (0020,000D)
patient_mrn	String (hashed)	Medical Record Number (privacy-preserved)	EHR linkage
extracted_entities	List<Entity>	Named entities (anatomy, pathology, dates)	NLP pipeline
embedding_vector	Float[768]	Dense representation for semantic search	BiomedBERT encoder

2.2.2 Agent 2: Clinical Historian (Structured Interview System)

Purpose: Transform unstructured patient input into clinically meaningful context through guided questioning.

State Transition — Clinical Interview Flow:

IDLE	User submits symptom note	Parse free text, extract chief complaint	CONTEXT_RETRIEVAL
CONTEXT_RETRIEVAL	Chief complaint identified	Query Knowledge Graph for related history	GUIDED_QUESTIONING
GUIDED_QUESTIONING	History retrieved	Generate targeted follow-up questions (max 5)	AWAITING_RESPONSE

AWAITING_RESPONSE	User responds	Integrate response, check completeness	COMPILE GUIDED_QUESTIONING
COMPILE	Interview complete	Generate Structured Clinical History document	OUTPUT

Output Schema — Structured Clinical History:

chief_complaint	Primary reason for imaging	Single sentence, SNOMED-coded
history_of_present_illness	Temporal narrative of symptom evolution	Structured timeline with dates
relevant_past_history	Prior imaging, surgeries, diagnoses	Auto-populated from Knowledge Graph
suspected_anatomical_correlation	Hypothesized affected structures	RadLex anatomical terms only
clinical_question	Specific question for the Reasoning Agent	Explicit, answerable query

2.3 Layer 2: Triage Layer

2.3.1 Agent 3: Frame Selector (Scout Agent)

Purpose: Reduce computational burden by identifying the subset of slices (frames) containing relevant pathology.

Model Architecture: EfficientFormer-L1 (~12M parameters) fine-tuned on BiomedCLIP embeddings

Target Performance: <2ms per slice encoding on RTX 4090/5090

Reduction Target: 95% slice reduction (500 slices → 10-25 key slices)

Selection Algorithm Logic Flow:

1	Text Encoding	Clinical question (string)	Query vector $Q \in \mathbb{R}^{512}$
2	Batch Slice Encoding	All slices $S_1 \dots S_n$	Visual vectors $V_1 \dots V_n \in \mathbb{R}^{512}$
3	Similarity Computation	$Q, V_1 \dots V_n$	Similarity scores $\text{sim}(Q, V_i)$
4	Temporal Smoothing	Raw similarity scores	Smoothed scores (window=5)
5	Peak Detection	Smoothed scores	Candidate indices
6	Anatomical Validation	Candidate indices	Validated Key Slice Manifest

2.4 Layer 3: Analysis Layer

2.4.1 Agent 4: Spatial Analyst (Reasoning Engine)

Purpose: Perform deep, multi-step reasoning over selected key slices to generate spatially-grounded findings.

Model Architecture: Qwen3-Medical-GRPO (70B parameters, Q4_K_M quantized)

Context Window: 128K tokens (sufficient for ~25 high-resolution slices + history)

Output Format: Structured XML with <think> reasoning traces and <answer> conclusions

2.4.2 Agent 5: Knowledge Verifier (Validation Engine)

Purpose: Ground AI-generated findings in established medical knowledge and detect hallucinations.

Primary Function: Cross-reference findings against RadLex ontology and clinical guidelines

Secondary Function: Calculate semantic distance between textual claims and visual features

2.4.3 Agent 6: Geometric Validator (Safety Gate)

Purpose: Enforce spatial consistency between textual claims and bounding box coordinates.

Type: Deterministic logic engine (non-neural)

Critical Role: Final gate before report generation; triggers Human-in-the-Loop on failure

Validation Rule Schema:

Rule ID	Check Name	Logic Expression	Action
GV-001	Containment Check	$\text{IoU}(\text{box}, \text{anatomy_mask}) < 0.5$	Raise SPATIAL_HALLUCINATION flag
GV-002	Exclusion Check	$\text{IoU}(\text{box}, \text{exclusion_mask}) > 0.3$	Raise ANATOMICAL_ERROR flag
GV-003	Size Plausibility	$\text{box_volume} \text{ outside } [\text{min}, \text{max}] \text{ for pathology type}$	Raise SIZE_ANOMALY flag

GV-004	Semantic Consistency	<code>cosine_dist(text_embed, visual_embed) > 0.4</code>	Raise DESCRIPTION_MISMATCH flag
GV-005	Temporal Consistency	Finding contradicts prior stable observation	Route to HUMAN_REVIEW

III. Reasoning Logic (ML Core)

3.1 Cognitive Architecture: System 2 Reasoning

MARO-v2 implements "System 2" reasoning as defined by Kahneman's dual-process theory. Unlike System 1 (fast, intuitive pattern matching), System 2 involves slow, deliberate, sequential analysis. This is achieved through Chain-of-Visual-Thought (CoVT) prompting enforced via Group Relative Policy Optimization (GRPO).

3.1.1 The MedVLM-R1 Reasoning Engine

Base Architecture: Qwen2.5-VL-72B-Instruct with medical domain adaptation

Training Paradigm: GRPO fine-tuning on reasoning-annotated medical VQA datasets

Key Innovation: Elimination of critic model enables 70B+ deployment on consumer hardware

3.2 Group Relative Policy Optimization (GRPO)

3.2.1 Mathematical Foundation

GRPO eliminates the memory overhead of traditional RLHF by computing advantages relative to a sampled group rather than a learned value function.

Optimization Objective:

$$\nabla_{\theta} J(\theta) = E[\sum_i (\hat{A}_i \cdot \nabla_{\theta} \log \pi_{\theta}(o_i|x)) - \beta \cdot \nabla_{\theta} DKL(\pi_{\theta} || \pi_{ref})]$$

Where:

- $\hat{A}_i$ = Advantage of output o_i , computed as: $(r_i - \mu(r)) / \sigma(r)$
- π_{θ} = Current policy (the model being trained)
- π_{ref} = Reference policy (frozen baseline to prevent drift)
- β = KL divergence penalty coefficient (prevents reward hacking)
- G = Group size (typically 4-8 sampled outputs per input)

3.2.2 Composite Medical Reward Function

The reward function $R(o)$ incentivizes structured reasoning, spatial grounding, and clinical accuracy:

$$R(o) = \lambda_1 \cdot R_{\text{format}} + \lambda_2 \cdot R_{\text{validity}} + \lambda_3 \cdot R_{\text{correctness}} + \lambda_4 \cdot R_{\text{spatial}} + \lambda_5 \cdot R_{\text{semantic}}$$

R_format	0.15	Enforces XML structure (<think>, <answer>, <box>)	Regex parsing; binary pass/fail
R_validity	0.20	Ensures clinical terminology validity	SNOMED-CT / RadLex lookup
R_correctness	0.25	Measures diagnostic accuracy	Label match or Model-as-Judge
R_spatial	0.25	Penalizes spatial hallucinations	IoU(predicted_box, ground_truth)
R_semantic	0.15	Visual-textual consistency	Cosine similarity of embeddings

3.3 Chain-of-Visual-Thought Protocol

3.3.1 The <think> Tag Protocol

All reasoning must be enclosed within <think> tags. This creates an auditable trace that can be inspected by the Validator and reviewed by clinicians.

Required Reasoning Phases:

Observation	<observe>	Raw visual description with coordinates	"Scanning RUL... opacity at [x:120, y:85, z:45]"
Hypothesis	<hypothesis>	Differential diagnosis list	"DDx: pneumonia, atelectasis, mass"
Evidence	<evidence>	Verification steps with boolean outcomes	"Air bronchograms: PRESENT. Silhouette sign: POSITIVE"
Synthesis	<synthesize>	Logical combination of evidence	"Air bronchograms + positive silhouette → consolidation"
Conclusion	</think>	Final diagnostic statement	"Lobar pneumonia, RUL"

3.3.2 Chain-of-Box Interleaving

MedGround-R1 extends CoVT by requiring bounding box coordinates to be interleaved with reasoning steps. This prevents the model from reasoning about anatomy it has not visually attended to.

Box Coordinate Schema:

x_min, x_max	Integer	Voxel index	Horizontal extent of bounding box
y_min, y_max	Integer	Voxel index	Vertical extent of bounding box
z_min, z_max	Integer	Slice index	Depth extent (through-plane)
confidence	Float [0,1]	Probability	Model's certainty in box placement
anatomy_label	String	RadLex term	Anatomical structure contained

3.4 Qwen3-Medical-GRPO Deployment Configuration

Base Model	Qwen2.5-VL-72B-Instruct	Best-in-class structured output generation
Quantization	Q4_K_M (GGUF)	Balances quality (>95% retention) with VRAM fit
Context Length	128K tokens	Accommodates 25 slices + full patient history
Inference Engine	llama.cpp with GPU offload	Native GGUF support, tensor parallelism
Vision Encoder	mmproj F16 (optional Q4)	Preserves spatial resolution for CoB
Sampling Strategy	Temperature 0.6, Top-P 0.9	Balances creativity with determinism

IV. Volumetric Vision (Data Science Core)

4.1 The Dimensionality Challenge

Standard Vision-Language Models process images as 2D grids. Medical imaging is inherently 3D (CT, MRI) or 4D (dynamic contrast). Processing volumetric data as a "bag of slices" destroys critical through-plane relationships—the continuity of vessels, airways, and organ boundaries.

4.2 RoMedFormer: The Visual Backbone

Architecture Class: 3D Vision Transformer with Rotary Positional Embeddings

Primary Innovation: Native volumetric tokenization preserving 3D spatial relationships

Secondary Innovation: Metal artifact resilience through transformer inpainting

4.2.1 Volumetric Tokenization

Unlike 2D ViTs that divide images into patches, RoMedFormer divides volumes into cubes (voxel patches). This preserves local 3D context at the embedding level.

Patch Size	16 × 16 × 4 voxels	Balances resolution with sequence length
Embedding Dimension	768	Standard transformer width
Number of Layers	24	Sufficient depth for complex anatomy
Attention Heads	12	Multi-head for diverse feature extraction
Pretraining Task	Masked Volume Reconstruction	Self-supervised; no labels required

4.2.2 Metal Artifact Resilience

Metal implants (pacemakers, hip replacements, surgical clips) cause "photon starvation" artifacts—dark streaks that obscure adjacent anatomy. RoMedFormer treats these as "corrupted tokens" and reconstructs them using global attention.

Performance Benchmarks:

PSNR (dB)	28.4	30.1	32.9	+9.3% vs ADN
SSIM	0.87	0.91	0.95	+4.4% vs ADN
NMAD	0.42	0.38	0.33	-13.2% vs ADN

4.3 Geometric-RoPE: DICOM-Calibrated Positional Encoding

4.3.1 The Anisotropy Problem

Medical scans are anisotropic: in-plane pixel spacing (e.g., 0.5mm) differs from slice thickness (e.g., 2.5mm). Standard 3D-RoPE treats all axes equally, distorting the model's understanding of true physical distance.

4.3.2 Physical Frequency Calibration

Geometric-RoPE scales rotation frequencies by the physical dimensions from DICOM metadata:

X (Width)	$\theta_x = \theta_{\text{base}} / \text{pixel_spacing}[0]$	High frequency for fine in-plane detail
Y (Height)	$\theta_y = \theta_{\text{base}} / \text{pixel_spacing}[1]$	High frequency for fine in-plane detail
Z (Depth)	$\theta_z = \theta_{\text{base}} / \text{slice_thickness}$	Low frequency for slow anatomical transitions

4.3.3 Frequency Splitting Strategy

The embedding dimension is partitioned into three chunks, each rotated independently by its respective axis:

d[0:256]	X (Width)	High	Horizontal boundaries, lateral extent
d[256:512]	Y (Height)	High	Vertical boundaries, superior/inferior
d[512:768]	Z (Depth)	Low	Through-plane continuity, organ extent

4.4 Anatomical Relative Indexing (ARI)

4.4.1 Anchor-Based Coordinate System

Absolute voxel coordinates are clinically meaningless because patient size and positioning vary. ARI anchors all observations to automatically-detected anatomical landmarks.

Chest	Carina (tracheal bifurcation)	RoMedFormer segmentation	Aortic arch
Abdomen	Celiac trunk origin	Vessel tracking	L1 vertebral body
Pelvis	Pubic symphysis	Bone detection	Sacral promontory
Spine	C2-C3 junction	Vertebral segmentation	Odontoid process

4.4.2 Relative Coordinate Schema

All findings are indexed relative to the primary anchor using physical millimeters:

anchor_structure	String (RadLex)	—	"RID1302" (Carina)
delta_x	Float	mm	+45.2 (right lateral)
delta_y	Float	mm	-12.8 (anterior)
delta_z	Float	mm	+85.0 (inferior/caudal)
confidence	Float [0,1]	—	0.94

Clinical Benefit: "Nodule at (+45, -12, +85)mm from Carina" is semantically meaningful and enables comparison across patients and timepoints, unlike "Nodule at voxel [256, 312, 89]"

V. Stateful Orchestration (Software Engineering)

5.1 LangGraph: The Orchestration Framework

MARO-v2 employs LangGraph to model the diagnostic workflow as a compiled state machine. Unlike conversation-based frameworks (e.g., AutoGen), LangGraph provides explicit control flow, typed state management, and—critically—checkpoint-based "Time Travel" for human correction.

5.1.1 Framework Selection Rationale

Control Flow	Dynamic (LLM-driven)	Explicit (Graph edges)	Determinism for clinical protocols
State Management	Implicit (chat history)	TypedDict schema	Precise variable tracking
Persistence	Session-based	Database checkpoints	Long-running workflows
Human Correction	Simple prompts	Time-Travel rollback	Error recovery without restart
Auditability	Log parsing	Native state history	Regulatory compliance

5.2 The Radiology State Schema

The application state is defined as a TypedDict that persists across the entire analysis lifecycle:

scan_id	str	Immutable	Unique identifier for the imaging study
patient_mrn	str (hashed)	Immutable	Privacy-preserved patient identifier
volume_paths	List[Path]	Immutable	Paths to preprocessed DICOM slices
clinical_history	ClinicalHistory	Immutable	Output from Clinical Historian agent

selected_indices	List[int]	Mutable (Triage)	Key slice indices from Scout Agent
visual_features	Dict[int, Tensor]	Mutable (Perception)	RoMedFormer embeddings per slice
reasoning_trace	List[str]	Mutable (Analysis)	Accumulated <think> blocks
findings	List[Finding]	Mutable (Analysis)	Structured findings with boxes
validation_errors	List[ValidationError]	Mutable (Validation)	Flags from Geometric Validator
human_corrections	List[Correction]	Mutable (HITL)	Radiologist override inputs
checkpoint_history	List[CheckpointID]	System-managed	IDs of all saved checkpoints

5.3 The Orchestration Graph

5.3.1 Node Definitions

N1	Ingestion Router	scan_id	volume_paths, clinical_history	Yes
N2	Scout Agent	volume_paths, clinical_history	selected_indices	Yes
N3	RoMedFormer	selected_indices, volume_paths	visual_features	No
N4	GraphRAG Retriever	patient_mrn, findings (partial)	temporal_context	No
N5	Reasoning Engine	visual_features, temporal_context, clinical_history	reasoning_trace, findings	Yes
N6	Geometric Validator	findings, visual_features	validation_errors	Yes
N7	Human Review	validation_errors, findings	human_corrections	Yes (Breakpoint)
N8	Report Generator	findings (validated), reasoning_trace	final_report	Yes

5.3.2 Edge Transition Logic

N1	N2	Always	Default
N2	N3	selected_indices not empty	Conditional
N2	N7	selected_indices empty (no relevant slices)	Error path
N3	N4	Always	Default
N4	N5	Always	Default
N5	N6	findings not empty	Conditional

N6	N7	validation_errors not empty	Error path
N6	N8	validation_errors empty	Success path
N7	N5	human_corrections contains RETRY	Cycle (retry reasoning)
N7	N2	human_corrections contains RE_TRIAGE	Cycle (re-select slices)
N7	N8	human_corrections contains OVERRIDE	Force continue

5.4 Time-Travel Debugging: The Human-in-the-Loop Protocol

5.4.1 Checkpointing Mechanism

Every node execution that modifies mutable state triggers a checkpoint save. Checkpoints are stored in a PostgreSQL database (or SQLite for development).

checkpoint_id	UUID	Unique identifier
thread_id	UUID	Groups checkpoints for a single analysis session
node_id	str	The node that created this checkpoint
timestamp	ISO8601	When the checkpoint was created
state_snapshot	JSONB	Complete serialized state
parent_checkpoint_id	UUID (nullable)	Previous checkpoint (for history traversal)

5.4.2 Time-Travel Workflow

Clinical Scenario: The Reasoning Engine incorrectly identifies a "Hilar Mass" that is actually an enlarged pulmonary artery.

1	Error detected at N6 (Validator)	Graph pauses; state saved to CP_6
2	Radiologist reviews reasoning_trace	Sees error in <think> block: vessel misidentified
3	Radiologist selects CP_5 (pre-Reasoning)	System loads state snapshot from CP_5
4	Radiologist injects hint into state	human_corrections.append("Structure is vascular, not mass")
5	System calls graph.update_state()	State modified in-place
6	Radiologist clicks "Resume"	Graph re-executes from N5 with corrected context

7	Reasoning Engine produces corrected finding	"Enlarged pulmonary artery consistent with pulmonary hypertension"
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5.4.3 Reflection Pattern

For low-confidence findings, MARO-v2 implements automatic reflection before escalating to human review:

confidence < 0.7	"Re-examine the borders. Is this a true mass or a summation artifact?"	Self-correction or confidence adjustment
multiple_ddx > 3	"Prioritize the differentials. Which has strongest visual evidence?"	Narrowed differential list
box_overlap > 0.5	"Your boxes overlap significantly. Clarify distinct findings."	Separated or merged findings

VI. Sovereign Knowledge Bridge (Data Engineering)

6.1 The Temporal Context Imperative

Radiology is inherently longitudinal. A finding's significance is determined by its temporal trajectory: a stable nodule is benign; a growing nodule is concerning. Standard AI systems process each scan in isolation, generating false positives and missing true progressions.

6.2 Architecture: Neo4j + ChromaDB Bridge

6.2.1 Dual-Store Design

Graph Store	Neo4j	Relationships, entities	Cypher traversal	Temporal queries, care pathways
Vector Store	ChromaDB	Embeddings	Cosine similarity	Semantic search, report retrieval
Bridge Layer	Custom adapter	UID mappings	Key lookup	Cross-referencing graph ↔ vector

6.2.2 Neo4j Schema (FHIR-Aligned)

Core Node Types:

:Patient	Patient	mrn_hash, birth_year	HAS_ENCOUNTER
:Encounter	Encounter	date, type, location	NEXT_ENCOUNTER, HAS_PROCEDURE
:Procedure	ImagingStudy	study_uid, modality	PRODUCED_OBSERVATION
:Observation	Observation	code, value, coordinates	CHANGED_FROM, REFERENCES_ANATOMY
:Anatomy	BodyStructure	radlex_id, name	CONTAINS, ADJACENT_TO
:Guideline	PlanDefinition	name, recommendation	TRIGGERED_BY

6.2.3 Temporal Edge Patterns

Time is modeled explicitly via relationship properties:

NEXT_ENCOUNTER	:Encounter	:Encounter	days_elapsed	Temporal sequence of visits
CHANGED_FROM	:Observation	:Observation	delta_size_mm, delta_density_hu	Finding evolution
TRIGGERED_BY	:Guideline	:Observation	match_criteria	When guideline applies

6.3 GraphRAG Query Patterns

6.3.1 Temporal Comparison Query

Query: "Has nodule A grown since last scan?"

Traversal Pattern:

```
MATCH (current:Observation {id: $nodule_id})-[:CHANGED_FROM*]->(prior:Observation)
RETURN prior.size_mm, prior.date ORDER BY prior.date DESC
```

Output: List of prior measurements with timestamps, enabling growth rate calculation.

6.3.2 Guideline Activation Query

Query: "What clinical action does this finding require?"

Traversal Pattern:

```
MATCH (obs:Observation {id: $finding_id})<-[:TRIGGERED_BY]-(g:Guideline) WHERE
obs.size_mm > g.threshold_mm RETURN g.name, g.recommendation
```

Output: Applicable guideline (e.g., "Fleischner 2017: Nodule >8mm requires PET-CT or biopsy").

6.4 ChromaDB Vector Index

6.4.1 Embedding Strategy

Radiology Reports	BiomedBERT	768	Section-level (Findings, Impression)
Clinical Notes	ClinicalBERT	768	Paragraph-level
DICOM Metadata	Custom encoder	256	Study-level aggregation

6.4.2 Bridge Mechanism

Every ChromaDB document stores a neo4j_node_id in its metadata. This enables seamless cross-referencing:

- Semantic query to ChromaDB: "Find reports mentioning lung nodules"
- Retrieve neo4j_node_ids from matching documents
- Execute Cypher query to fetch temporal relationships
- Return combined semantic + structural context to Reasoning Engine

VII. Infrastructure & Deployment (DevOps)

7.1 Hardware Architecture

MARO-v2 is designed for sovereign, on-premise deployment. Patient data never leaves the hospital network. This section defines the minimum and recommended hardware configurations.

7.1.1 GPU Selection: RTX 5090 vs. Dual RTX 4090

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VRAM Total	32 GB GDDR7	48 GB GDDR6X	70B Q4 requires ~42GB; 5090 insufficient alone
Memory Bandwidth	1,792 GB/s	~1,008 GB/s (per card)	5090 is 78% faster for models that fit
Interconnect	PCIe 5.0 x16	PCIe 4.0 x16 (each)	No NVLink; tensor parallelism via PCIe
Power Draw	575W TDP	900W combined TDP	Dual 4090 requires robust PSU (1200W+)
Estimated Cost	\$2,000–\$2,600	\$3,600 (for pair)	Dual 4090 is 40% more expensive
Optimal Role	Scout Agent, Validator	Reasoning Engine (70B)	Hybrid deployment recommended

7.1.2 Recommended Server Configuration

CPU	AMD EPYC 7313 (16C)	AMD EPYC 7543 (32C)	For preprocessing and orchestration
RAM	128 GB DDR4 ECC	256 GB DDR5 ECC	For large DICOM volumes
GPU (Reasoning)	Dual RTX 4090	Dual RTX 5090 (when available)	Primary inference target
GPU (Triage)	Single RTX 4090	Single RTX 5090	High-throughput Scout encoding
Storage (Fast)	2 TB NVMe Gen4	4 TB NVMe Gen5	Active study cache
Storage (Archive)	10 TB RAID-6 HDD	100 TB SAN	Long-term PACS integration
Network	10 GbE	25 GbE	For PACS pull/push operations

7.2 Quantization Strategy

7.2.1 AWQ vs. GGUF Comparison

Inference Speed	Fastest (vLLM optimized)	Fast (llama.cpp)	AWQ → Scout; GGUF → Reasoner
Quality Retention	~92–95%	~95–97%	GGUF preferred for reasoning
Stability	Occasional instability	Highly stable	GGUF for critical path
GPU Offload	Full GPU only	Partial CPU offload possible	GGUF more flexible
Vision Encoder	FP16 only	Q4 or FP16	GGUF more space-efficient

7.2.2 VRAM Budget Allocation (Dual RTX 4090, 48GB)

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Qwen3-Medical-72B (Policy)	Q4_K_M	38.0	79%
Vision Encoder (mmproj)	F16	2.5	5%
KV Cache (128K context)	—	4.0	8%
CUDA Overhead	—	2.0	4%
Safety Buffer	—	1.5	3%
TOTAL	—	48.0	100%

7.3 Deployment Architecture

7.3.1 Container Strategy

Orchestrator (LangGraph)	maro-orchestrator:v2	None	2 (HA)
Scout Agent	maro-scout:v2	1x RTX 5090	1
Reasoning Engine	maro-reasoner:v2	2x RTX 4090	1
Geometric Validator	maro-validator:v2	None (CPU)	2
Neo4j (Graph)	neo4j:5.x-enterprise	None	3 (cluster)
ChromaDB (Vector)	chromadb:0.5.x	None	2 (HA)
PostgreSQL (Checkpoints)	postgres:16	None	2 (primary-replica)

7.3.2 Latency Budget

DICOM Ingestion (500 slices)	< 5s	3.2s	I/O bound
Scout Triage (500 slices)	< 2s	1.1s	GPU compute
RoMedFormer Encoding (25 slices)	< 3s	2.4s	GPU compute
GraphRAG Retrieval	< 500ms	280ms	Database query
Reasoning (25 slices)	< 45s	38s	LLM inference
Geometric Validation	< 1s	0.6s	CPU compute
TOTAL (excluding human review)	< 60s	~46s	—

VIII. Appendices

Appendix A: Glossary of Terms

ARI	Anatomical Relative Indexing — coordinate system anchored to anatomical landmarks
AWQ	Activation-aware Weight Quantization — 4-bit quantization optimized for speed
CoB	Chain-of-Box — prompting technique interleaving bounding boxes with reasoning
CoVT	Chain-of-Visual-Thought — visual reasoning protocol with explicit observation steps
GGUF	GPT-Generated Unified Format — flexible quantization format for llama.cpp
GRPO	Group Relative Policy Optimization — RL algorithm without critic model
HITL	Human-in-the-Loop — clinical workflow requiring human verification
IoU	Intersection over Union — metric for bounding box overlap
LangGraph	State machine framework for agentic AI workflows
RoPE	Rotary Positional Embedding — position encoding via vector rotation
TKG	Temporal Knowledge Graph — graph database with explicit time modeling

Appendix B: Document Revision History

1.0	2025-10-15	Research Team	Initial vision document (5-agent mesh)
1.5	2025-12-01	ML Engineering	V1 Research: GRPO integration, CoVT protocol
2.0-DRAFT	2026-01-20	Architecture Team	V2 Research: LangGraph, 3D-RoPE, hardware specs
2.0-FINAL	2026-02-01	Full Panel	SDD consolidation; authoritative source of truth

Appendix C: Compliance & Regulatory Notes

MARO-v2 is designed as a Clinical Decision Support (CDS) system under FDA guidance. It is explicitly NOT intended for autonomous diagnosis. All outputs require clinician verification.

- FDA Classification Target: Class II (510(k) pathway)
- HIPAA Compliance: All patient identifiers hashed; no PHI in model weights
- Audit Trail: Complete state history preserved via LangGraph checkpoints
- Mandatory Disclaimer: All reports include non-diagnostic advisory language

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